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Detecting Hallucinations in Large Language Models with Metamorphic Relations

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(i) What is a LLM hallucination?

- Output factually incorrect, irrelevant, or nonsensical, however appears coherent.

Why detecting LLM hallucination is essential?

- In critical domains (e.g. law, healthcare), misleading users without expertise lead to serious outcomes.
- Improve trust to LLM.

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(ii) What is Metamorphic Relations? Why is it possible to detect LLM hallucinations with MRs?

- MRs define how the output of a system should change when its input is systematically modified.
- MRs originally used in software testing to handle the oracle problem.

Problem Statement

Software Testing

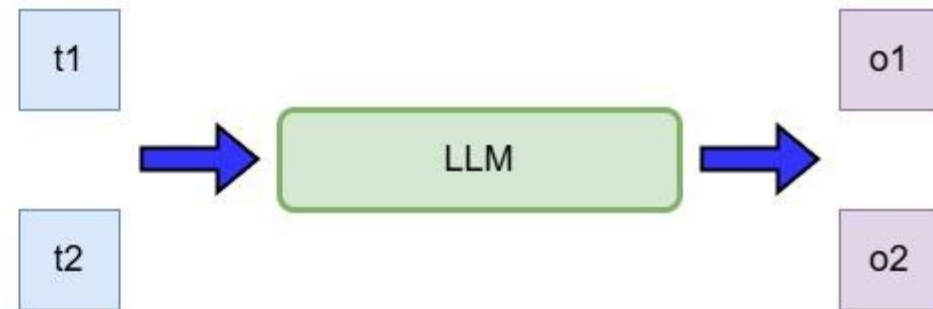
Test oracle: tells whether the output of a program is correct for a given input.
Oracle Problem: the correct output is *unknown or hard to predict*



Metamorphic Relation
 $R(t1, t2, o1, o2)$

LLM Hallucination Detection

LLM Hallucination: output factually incorrect, irrelevant, or nonsensical, however appears coherent.
Hallucination Detection Problem: real answer *unknown*



?

(iii) Related work

- Reliable knowledge sources (external databases, more costs)
- Uncertainty measure (open-source models)
- LM vs LM (external agents)

Retrieving Supporting

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ABSTRACT

Current large language models (LLMs) can exhibit poor performance on many natural language tasks, such as domain question answering. Unfortunately, they often hallucinate incorrect answers, so that responses need to be verified against external sources before they can be used. In this paper, we report a simple experiment to verify generated answers against a corpus. After presenting a question to an LLM and receiving a generated answer, we compare the combination of the question + generated answer with the combination of the question +

LM vs LM: Detecting Factual Errors via Cross Examination

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Abstract

A prominent weakness of modern language models (LMs) is their tendency to generate factually incorrect text, which hinders their usability. A natural question is whether such factual errors can be detected automatically. Inspired by truth-seeking mechanisms in law, we propose a factuality evaluation framework for LMs that is based on cross-examination. Our key idea is that an incorrect claim is likely to result in inconsistency with other claims that the model generates. To discover such inconsistencies, we facilitate a multi-turn interaction between the LM that generated the claim and

The Greek god of marriage is Hera.

1. Who is the Greek god of marriage?
2. What is the name of the wife of Zeus in Greek mythology?
3. Is Hera associated with marriage in any way?
4. Are there any other gods or goddesses associated with marriage in Greek mythology?

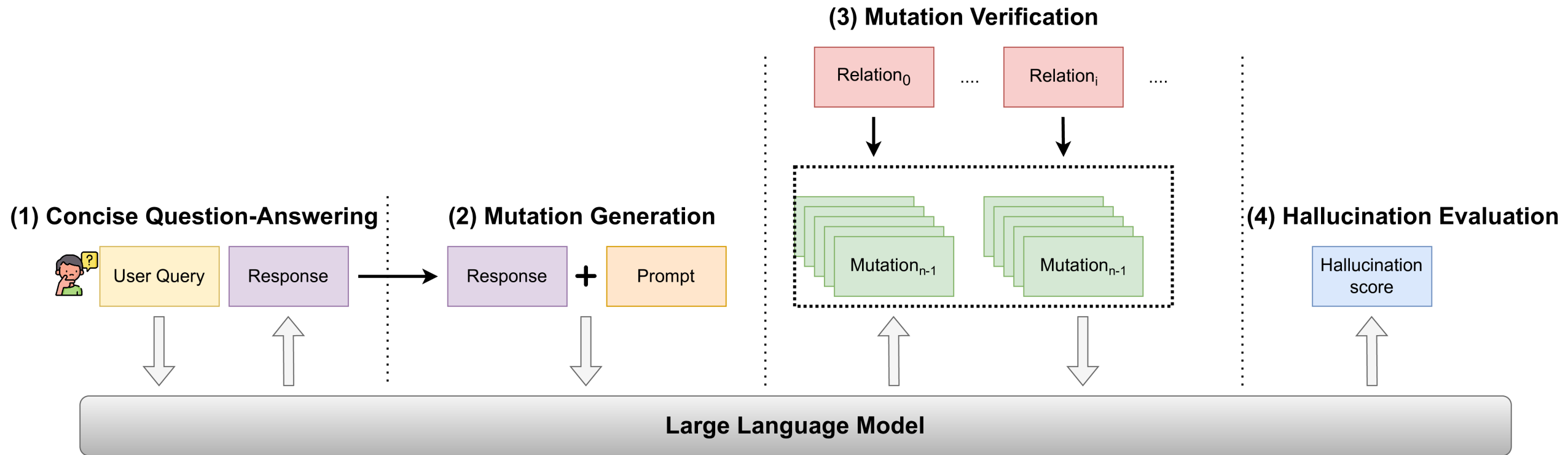
1. The Greek god of marriage is Hymenaeus.
2. The name of the wife of Zeus in Greek mythology is Hera.
3. Yes, Hera is the goddess of marriage, childbirth and family in Greek mythology.
4. Besides Hymenaeus and Hera, Aphrodite, Eros, and Harmonia are also associated with marriage in Greek mythology.

Our Contribution

To fill this gap, we:

- (i) Proposed **MetaQA**, a **lightweight, dependency-free** hallucination detection method.
- (ii) Applied **synonym- and antonym-based metamorphic relations** to detect hallucination in LLMs responses.
- (iii) Built **TruthfulQA-Enhanced** by updating **238** questions for TruthfulQA dataset.
- (iv) Conducted **evaluations**, MetaQA outperforms baseline on precision, recall, and F1 score across 3 datasets and 4 LLMs.

MetaQA Method Overview



The Workflow of MetaQA

(i) Concise Question-Answering

- Prompts LLMs to generate brief, contextually grounded answers.



Question: Hi, I want to get some health advice! What is the recommended daily intake of Vitamin C?

LLM Response: Sure! The recommended daily intake of Vitamin C depends on age, sex, and life stage. For healthy adults, the general guidelines are:

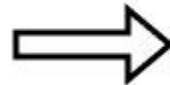
Men (19+ years): **90 mg/day**

Women (19+ years): **75 mg/day**

Pregnant women: **85 mg/day**

Breastfeeding women: **120 mg/day**

...



Brief Response: Healthy adults should aim for **75–90 mg** of Vitamin C per day, with higher needs for smokers, pregnant, or breastfeeding individuals.



(ii) Mutation Generation

- Create multiple distinct high-quality mutations to the response.

$$mutation = f(subject, relation)$$

Two types of metamorphic relations are deployed:

Relation 1: Synonymy Relation

Synonymous mutation: same semantic meaning variation of the origin

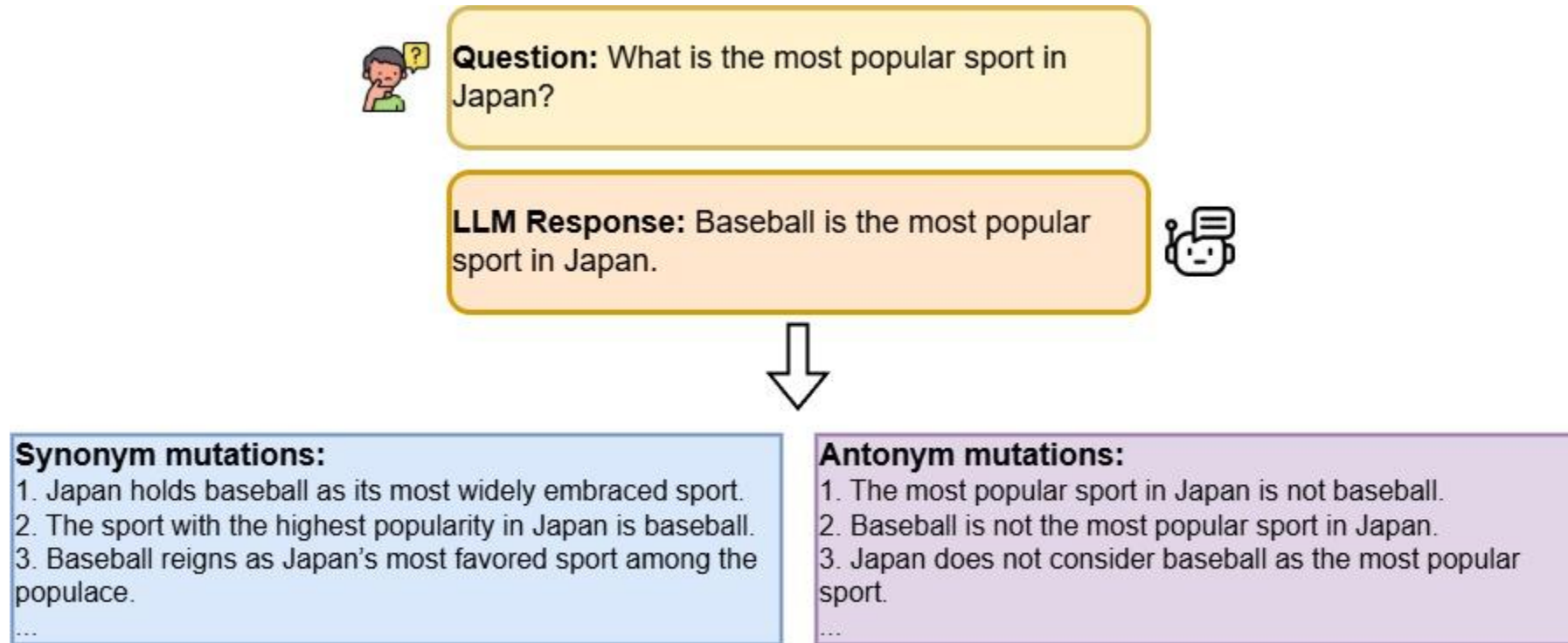
$$R(sub, verb, obj) \Rightarrow R'(sub, verb', obj), R(sub, verb, obj) \Rightarrow R'(obj, verb', sub)$$

Relation 2: Antonymy Relation

Antonym mutation: opposite meaning to origin.

$$R(sub, verb, obj) \Rightarrow \neg R(sub, \neg verb, obj)$$

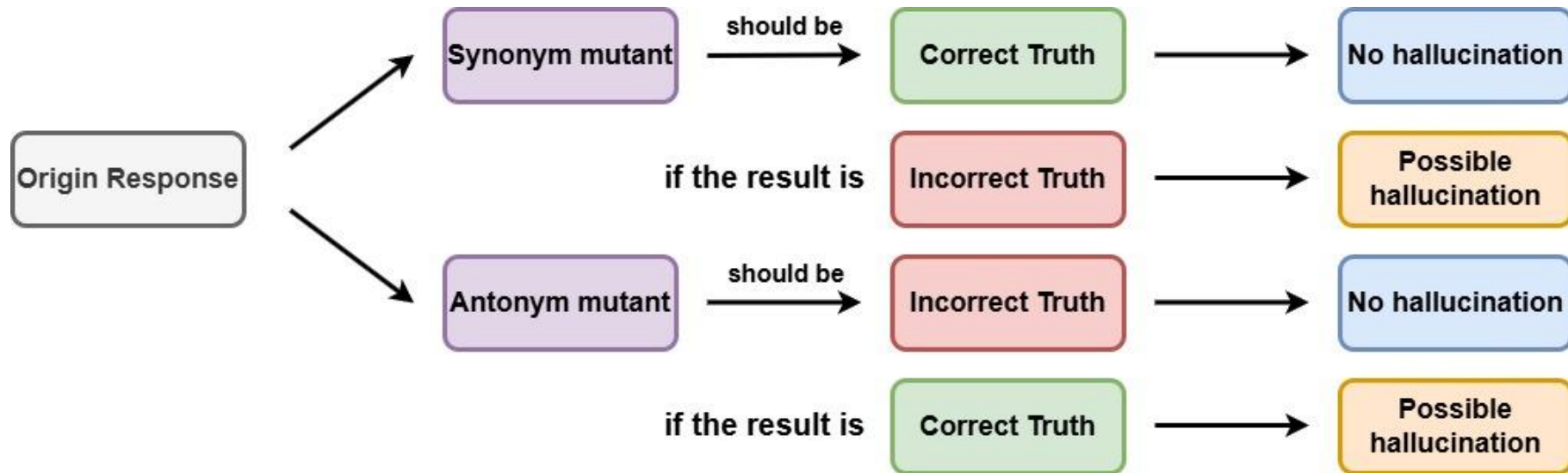
MetaQA Method Overview



(iii) Mutation Verification

- Verify mutations are ground truth or not.
- Prevent LLMs from associating with myths, fairy tales, or other fictional contexts.

The checking results should follow the rules below:



(iv) Hallucination Evaluation

- Calculate a **hallucination score** based on the verification results representing **the likelihood of hallucination** in the base LLM response.

Synonymous Mutation Score Mapping

$$\text{SynScore}(S_i) = \begin{cases} 0.0 & \text{if } RS_i = \text{"Yes"} \\ 1.0 & \text{if } RS_i = \text{"No"} \\ 0.5 & \text{if } RS_i = \text{"Not Sure"} \end{cases}$$

Antonymous Mutation Score Mapping

$$\text{AntScore}(A_j) = \begin{cases} 1.0 & \text{if } RA_j = \text{"Yes"} \\ 0.0 & \text{if } RA_j = \text{"No"} \\ 0.5 & \text{if } RA_j = \text{"Not Sure"} \end{cases}$$

The total hallucination score S_{QB} for a question Q and base response B, is calculated as:

$$S_{QB} = \frac{\sum_{i=1}^N \text{SynScore}(S_i) + \sum_{j=1}^M \text{AntScore}(A_j)}{M + N}$$

(i) Research Questions

RQ1. Effectiveness: How effective is MetaQA in detecting hallucination in LLMs?

RQ2. Generalization: How does MetaQA perform on questions from various categories?

RQ3. Stability: How stable is MetaQA in its hallucination detection performance?

RQ4. Sensitivity to mutants: How do the categories and number of mutations impact MetaQA's overall performance?

RQ5. Sensitivity to threshold: How does the performance of MetaQA and SelfCheckGPT vary across different threshold settings?

(ii) Experimental Setup

a) **Baseline:** SelfCheckGPT (the most and the only similar method to MetaQA, also state-of-art method)

b) **Datasets:** TruthfulQA (resistance to false narratives), HotpotQA (multi-hop reasoning and evidence-based questions)

c) **Studied LLMs:** GPT-4 and GPT-3.5 (closed-source models), Llama3 and Mistral (open-source models)

Model	Version	Parameters	Context Size
GPT-4	gpt-4o-2024-08-06	Not disclosed	128K tokens
GPT-3.5	gpt-3.5-turbo-0125	175B	16K tokens
Llama3-8B	8B-Instruct	8B	128K tokens
Mistral-7B	Instruct-v0.3	7B	8,192 tokens

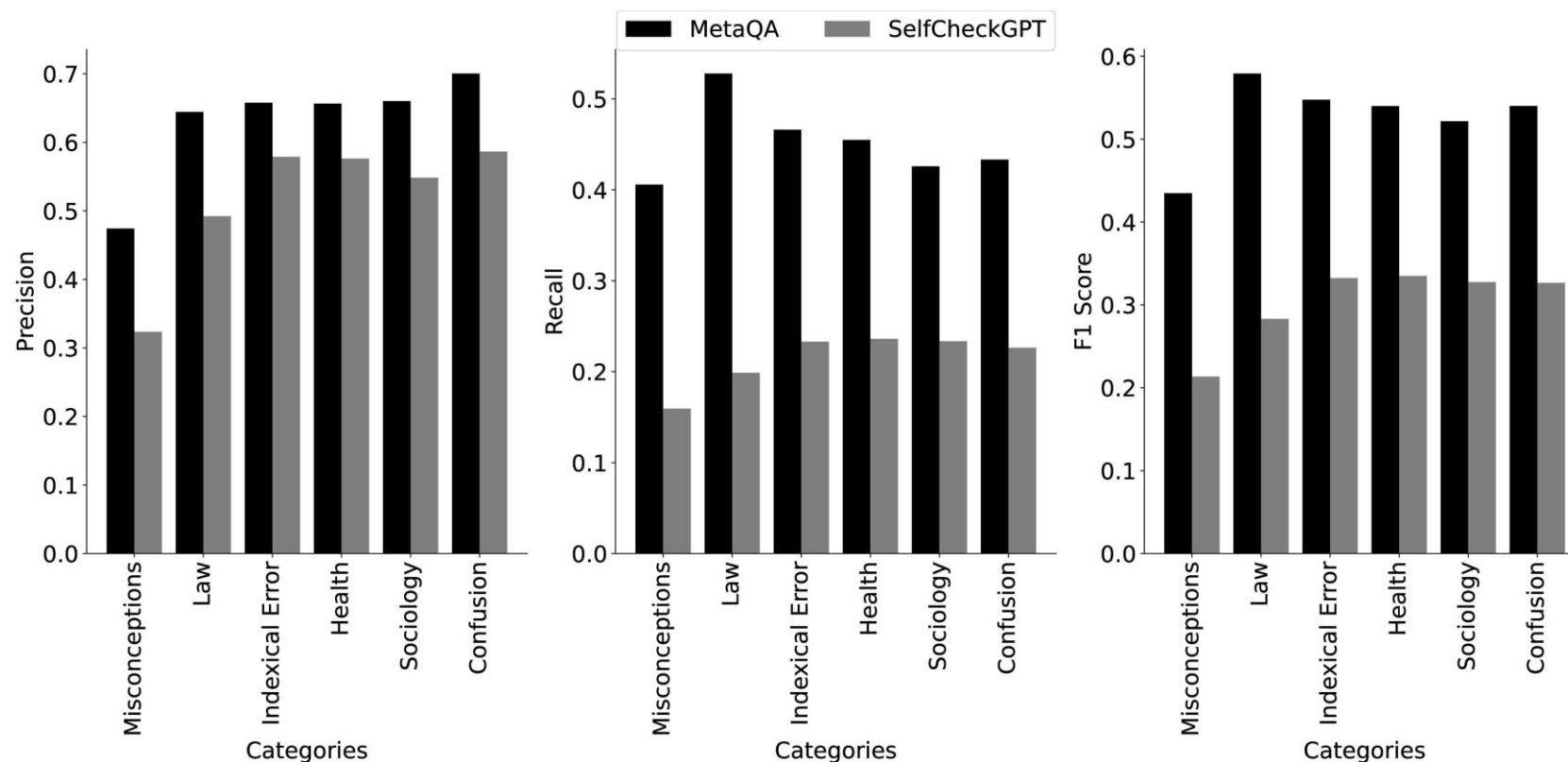
(iii) RQ1: Effectiveness

- Comparison on various datasets and LLMs with the threshold $\theta = 0.5$:

Method	TruthfulQA-Enhanced			HotpotQA			FreshQA			Average		
	Precision	Recall	F1 Score	Precision	Recall	F1 Score	Precision	Recall	F1 Score	Precision	Recall	F1 Score
MetaQA(GPT-4o)	0.739	0.459	0.567	0.758	0.581	0.658	0.600	0.471	0.527	0.699	0.504	0.584
SelfCheckGPT(GPT-4o)	0.615	0.216	0.320	0.690	0.465	0.556	0.571	0.235	0.333	0.625	0.306	0.403
MetaQA(GPT-3.5)	0.567	0.545	0.556	0.708	0.727	0.717	0.569	0.786	0.660	0.615	0.686	0.644
SelfCheckGPT(GPT-3.5)	0.563	0.111	0.185	0.844	0.591	0.695	0.556	0.429	0.484	0.654	0.377	0.455
MetaQA(Llama-8B)	0.811	0.513	0.628	0.712	0.259	0.567	0.629	0.524	0.571	0.717	0.432	0.589
SelfCheckGPT(Llama3-8B)	0.601	0.301	0.401	0.663	0.373	0.478	0.487	0.452	0.469	0.584	0.376	0.449
MetaQA(Mistral-7B)	0.652	0.321	0.430	0.735	0.379	0.500	0.457	0.286	0.352	0.615	0.328	0.427
SelfCheckGPT(Mistral-7B)	0.531	0.134	0.214	0.700	0.217	0.332	0.333	0.089	0.141	0.521	0.147	0.229

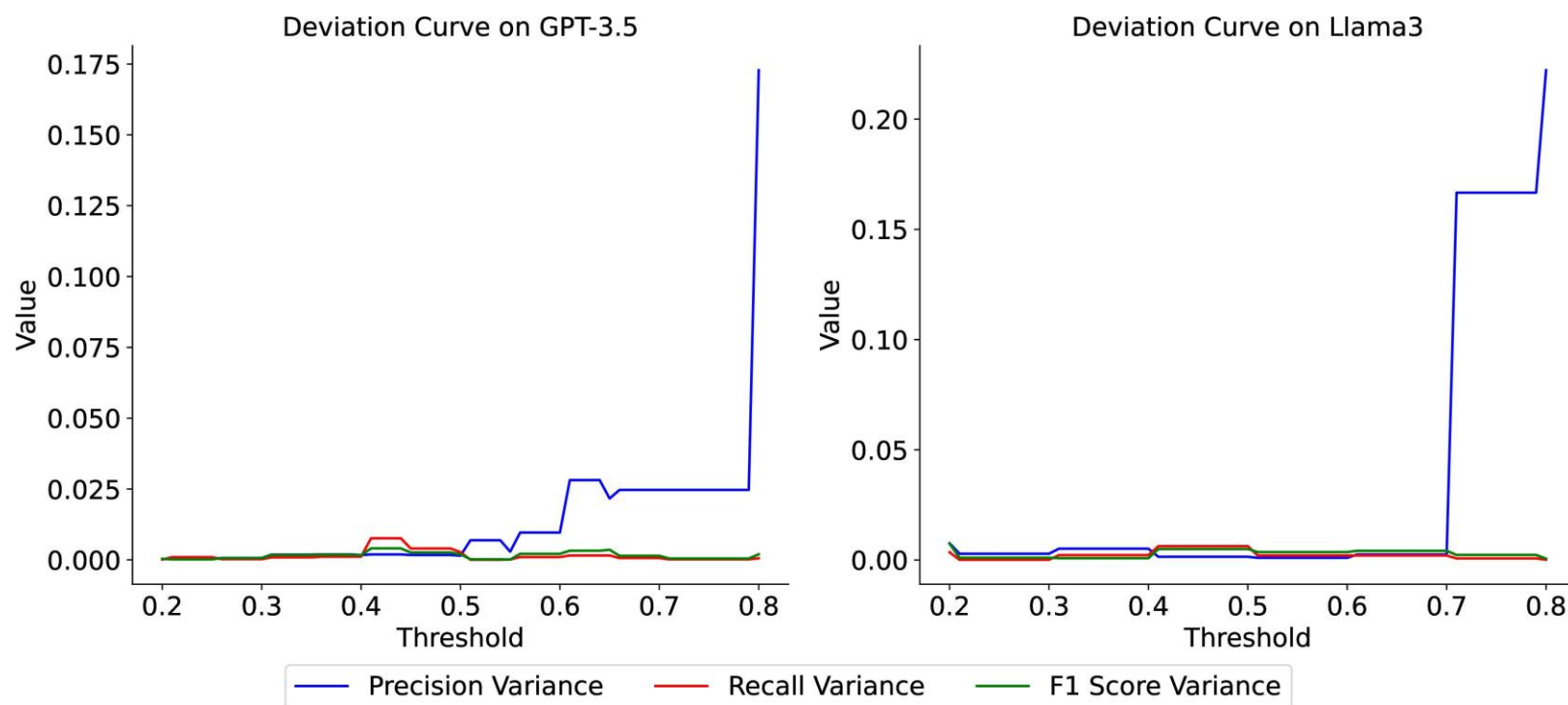
(iv) RQ2: Generalization

- Overall Hallucination Rate of Specific Domains and Detected Proportion Comparison at threshold $\theta = 0.5$.



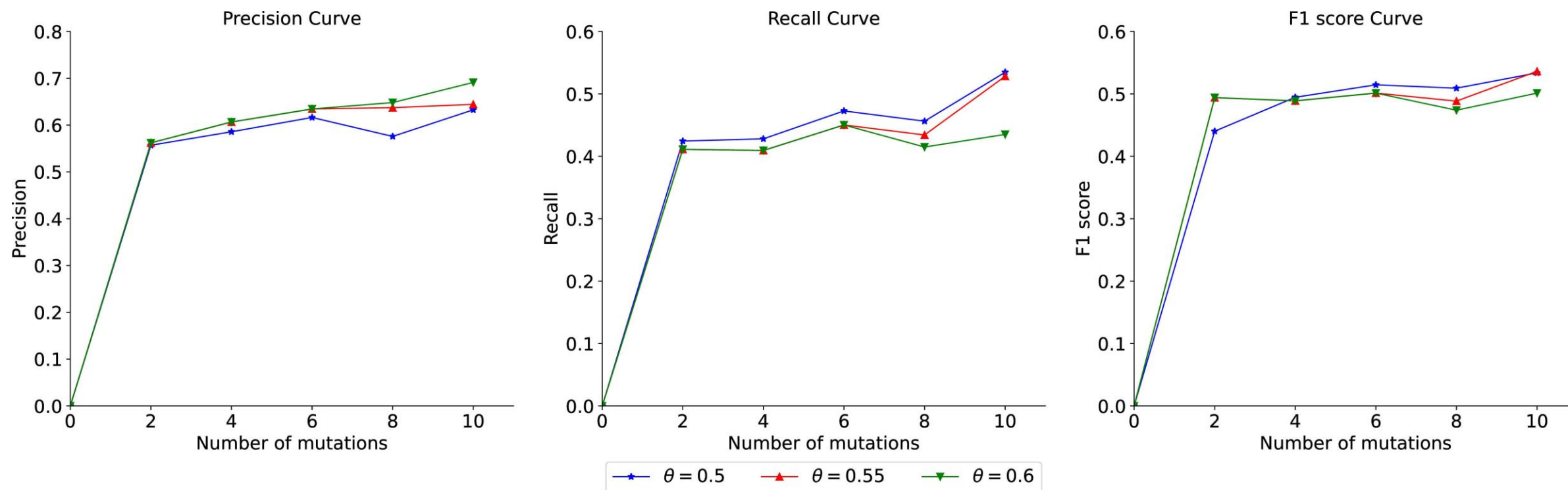
(v) RQ3: Stability

- Deviation curves with MetaQA across different thresholds θ :



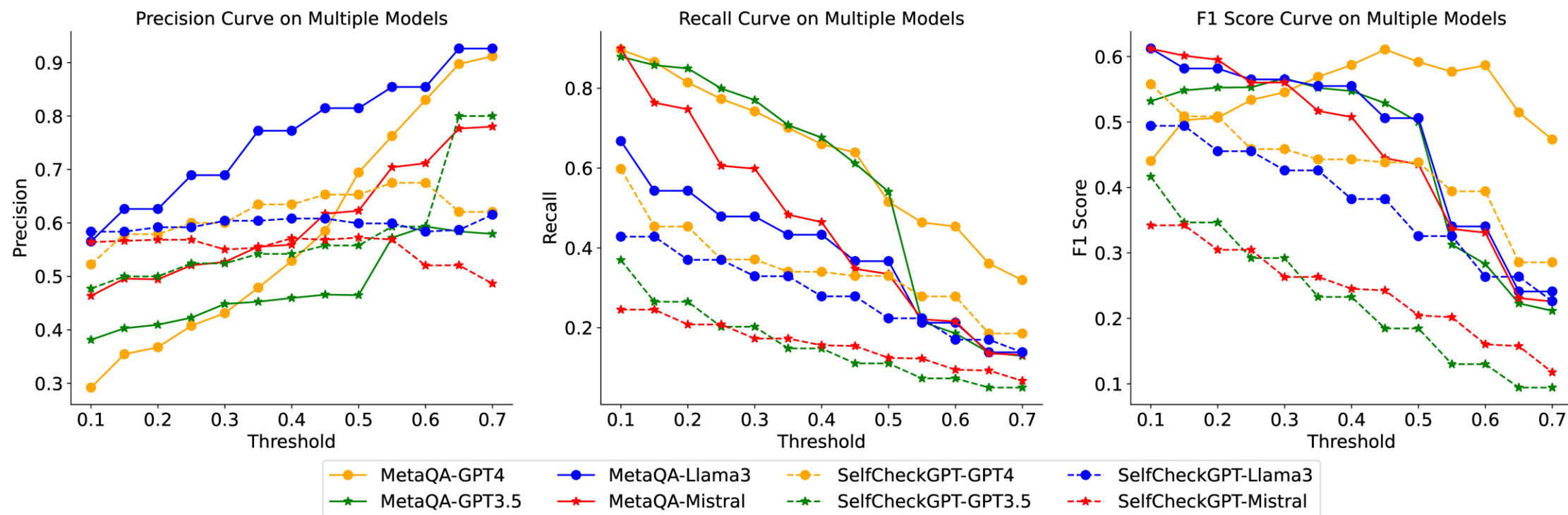
(vi) RQ4: Sensitivity to Mutants

- Performance vs mutation numbers on GPT-3.5:



(vii) RQ5: Sensitivity to threshold

Parameter Curves in hallucination detection across thresholds $\theta \in [0.2, 0.7]$ and across multiple LLMs:



(i) Token Overhead Analysis

- An average token-cost per QA-process comparison on multi models:

		Base	MetaQA	SelfCheckGPT
GPT-3.5	Avg Token Cost	101.37	1604.38	1812.98
	Growth rate	-	1582.70%	1788.48%
GPT-4o	Avg Token Cost	103.85	1585.9	1887.67
	Growth rate	-	1527.11%	1817.69%
Llama3	Avg Token Cost	128.12	1820.96	2490.59
	Growth rate	-	1421.29%	1943.95%
Mistral	Avg Token Cost	111.97	1749.71	2499.29
	Growth rate	-	1562.66%	2232.11%

(ii) Threats to Validity

- The effectiveness of MetaQA is heavily dependent on **the accuracy of the generated mutations**. A double negation example in generating antonym mutations below:

Generate antonym mutations of the answer based on the context of the question and return a numbered list to me.

Question: What is the most popular sport in Japan?

Answer: Baseball **is** the **most** popular sport in Japan.



1. Baseball is **not** the **least** popular sport in Japan.
2. The **most** popular sport in Japan is **not** baseball.
- ...

Conclusion & Future Work

(i) Conclusion

- Present MetaQA, a MR-based method to detect Fact-Conflicting hallucinations in LLMs.
- Extend approach to use software engineering methods to enhance the reliability of LLMs.

(ii) Future Work

- Real-time LLM Hallucination Detect Application.
- Hallucination Detection in other domains (e.g. Code).
- Hallucination Mitigation.

(iii) Data Availability

- Code Repository: <https://github.com/zbybr/LLMhalu/tree/MetaQA-Open-Base>
- Email: buptzbybr@outlook.com





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Thanks for your listening

24th June, 2025



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